TECHNICAL SERVICE BULLETIN
Various Concerns Due To C115 Wiring Connector Water Ingress

26-2212
05 May 2026

Model:

Ford 2022-2026 Next-Generation Ranger
2022-2026 Ranger Raptor
2022-2026 Next-Generation Everest

Issue: Some 2022-2026 Next-Generation Ranger, Ranger Raptor and Everest vehicles may experience one or more of the following symptoms. This may be due to water ingress into the C115 wiring connector. The vehicle may exhibit any of the following symptoms:

- Power cut off to engine
- Not able to accelerate the vehicle or change gear
- Multiple illuminated Malfunction Indicator Lamps (MIL)/Multiple warnings on the Instrument Panel Cluster (IPC)
  - Fuse 10 blown in the Power Distribution Box (PDB)
  - Vehicle not turning on in next key cycle once turned off
  - Electronic Stability Control (ESC) service required
  - Gear shifter not working
  - Front camera not working
  - Pre collision assist malfunction
  - Service vehicle soon
  - Parking sensor malfunction
- Battery Drain
- Charging warning
- Horn keeps blowing
- Fuse 9 blown in the PDB
- Front camera cut off for 2 seconds
- IPC not working
- Air Conditioning (A/C) not working
- Light switch not working (headlamps and side indicators)
- Hazard lights and horn not working
- Fuse 25 blown in the PDB
- Fuse 101 blown in the PDB
- Right Hand Side (RHS) headlamp turned off

Action: Should a vehicle present with the above concern, Technicians are advised to follow the Service Procedure below.

Parts

Service Part Number	Claim Quantity	Package Order Quantity	Number in Package	Description
RB3Z14S411HVB		1	1	Kit - Wiring Assembly
RB3Z14S411VBB		1	1	Kit - Wiring Assembly
N1WZ14F662F		1	1	Cable Assembly
N1WZ14F662R		1	1	Cable Assembly
RB3Z14S411HWA		1	1	Kit Wiring
RB3Z14S411VCA		1	1	Kit Wiring

Labor Times

Description	Operation No.	Time
Vehicle fitted with front camera repair wiring up to 40 wiring pins	262212A	10.3 Hrs.
Vehicle fitted with front camera repair wiring up to 56 wiring pins	262212B	11.3 Hrs.
Vehicle without front camera repair wiring up to 40 wiring pins	262212C	8.8 Hrs.
Vehicle without front camera repair wiring up to 56 wiring pins	262212D	9.8 Hrs.

Repair/Claim Coding

Causal Part:	14A005
Condition Code:	x4

PDR Table

PDR Code	Description	Time
TSB262212A	Vehicle fitted with front camera repair wiring up to 40 wiring pins	10.3 Hrs.
TSB262212B	Vehicle fitted with front camera repair wiring up to 56 wiring pins	11.3 Hrs.
TSB262212C	Vehicle without front camera repair wiring up to 40 wiring pins	8.8 Hrs.
TSB262212D	Vehicle without front camera repair wiring up to 56 wiring pins	9.8 Hrs.

Check and Record the Diagnostic Trouble Codes (DTCs)

1. Connect the Ford Diagnosis and Repair System (FDRS) and check for DTCs.

(1). Check for DTCs and record all stored DTCs on the Repair Order (RO). The recorded DTCs are required later in the Service Procedure.

(2). Clear the DTCs.

2. Are any of the symptoms referenced in this TSB present on the vehicle and/or do any of the DTCs recorded at Step 1 relate directly to the symptoms referenced in this TSB?

(1). Yes - proceed to the next step (C115 Connector Inspection) below.

(2). No - this TSB does not apply. Refer to the Workshop Manual (WSM) for diagnosis/repair.

C115 Connector Inspection

1. Inspect the C115 connector for water/moisture ingress. (Confirm: Check continuity of the circuit and inspect for pin corrosion or terminal pull-out issues (refer to Figure 1 and 2 below) (refer to the Professional Technician System (PTS), Wiring Tab, Cell 012, Page 1 Charging System).

WARNING: Always disconnect the battery ground strap before commencing work on the electrical system, to avoid damage to the electrical system. Refer to WSM Section 414-01.

2. Disconnect the battery. Refer to WSM Section 414-01.

3. Remove the Left Hand (LH) front wheel. Refer to WSM Section 204-04A.

4. Remove the LH front fender splash shield. Refer to WSM Section 501-02.

5. Working in the LH front wheelhouse, carefully disconnect the C115 connector.

6. With the use of a bright workshop lamp, carefully inspect both sides of the C115 connector for any signs of moisture entry and/or corrosion damage. (Figure 1 and 2)

NOTE: Figure 1 and 2 are a guide only. In some cases, the damage to the C115 connector may not be as immediately obvious as that shown in the illustrations, careful and thorough inspection is recommended. Under no circumstances should the C115 connector be disassembled any further at this stage of the TSB. Further disassembly of the C115 connector can disturb or damage the connector including the critical mat seal inside the connector and create a new water leakage pathway into the connector.

Figure 1

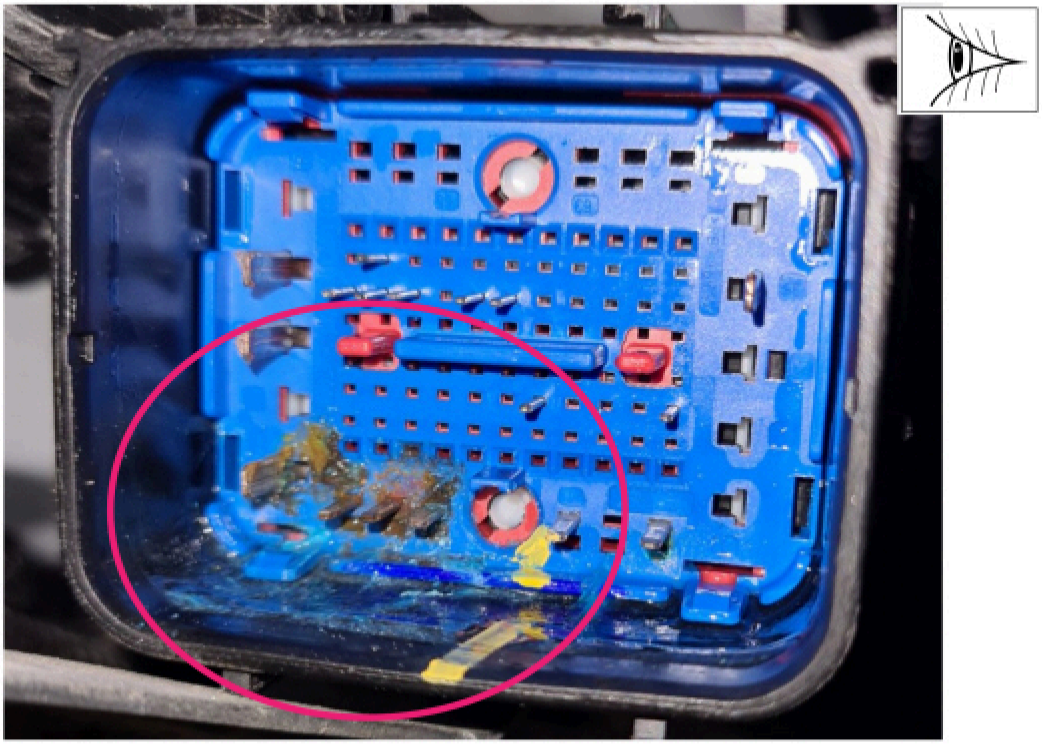

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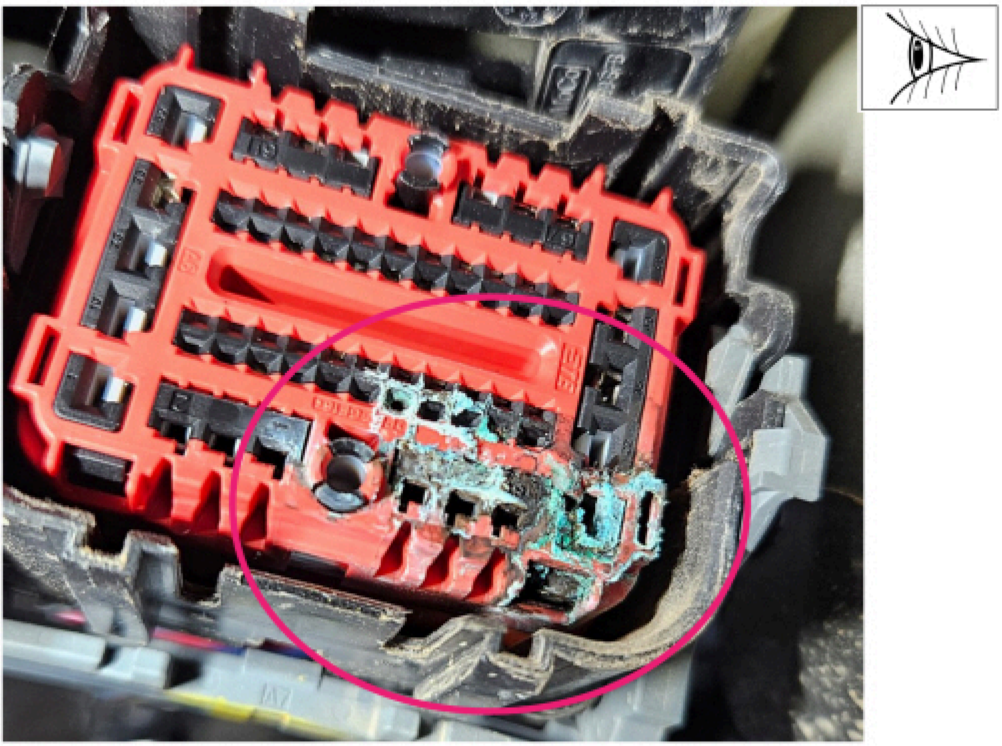
Figure 2



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7. Is there any water/moisture ingress or corrosion evident on the C115 connector?

(1). Yes – proceed to the next step (C115 Connector Replacement Procedure).

(2). No – this TSB does not apply. Refer to the WSM for further diagnosis/repair.

C115 Connector Replacement Procedure

This TSB is presented in the following sections:

- Removal of Surrounding Components in Preparation for the C115 Pigtail Repair
- Removal of Existing Harness Protective Coverings in Preparation for the C115 Pigtail Repair
- C115 Pigtail Connector Repair - General Warnings, Cautions and Repair Guidelines
- Reference Images of C115 Connector Layout

- Service Pigtail Kit Contents
- Recommended tools for the repair
- Reference table for wire color/sizes on TSB pigtail connector vs potential wire color/sizes on vehicle
- Wire Crimping Method
- C115 Pigtail Connector Installation
- Coaxial Overlay Installation – Engine Bay Side
- Coaxial Overlay Installation – Body Side
- Re-Installation of Surrounding Components Post Wiring Pigtail Repair and Post Repair Checks

IMPORTANT: The procedure provided in this TSB to cut and splice in the C115 pigtail and coaxial overlay is a complex repair and requires a very careful and methodical approach. Ensure you take time to carefully read the TSB in full and comprehend each step of the procedures in this TSB before starting the repair on the vehicle.

Removal of Surrounding Components in Preparation for the Wiring Pigtail Repair

WARNING: Always disconnect the battery ground strap before commencing work on the electrical system, to avoid damage to the electrical system. Refer to WSM Section 414-01.

1. Take clear photos of the corrosion damage on the C115 connector and save these for later attachment to the warranty claim.

2. Remove the battery. Refer to WSM Section 414-01.

3. Remove Body Control Module C (BCMC). Refer to WSM Section 419-10.

4. Remove the BCMC lower tray.

5. Disconnect the anti-theft alarm horn (optional step, vehicle feature dependent). Refer to WSM Section 419-01A.

6. Disconnect the wiring harness earth cable located under the BCMC. Refer to PTS, Wiring, Cell 012, Page 1 Charging System.

7. Disconnect the earth cable G104. Refer to PTS, Wiring, Cell 012, Page 1 Charging System.

8. Detach the wiring harness retaining clips.

9. Raise the vehicle. Refer to WSM Section 100-02.

10. Working in the LH wheelhouse, Detach the relay harness. Held on with a nut at the top and 2 clips at the bottom.

11. Disconnect and detach the C115 connector as well as the adjacent connectors. (Figure 3 and 4)

Figure 3

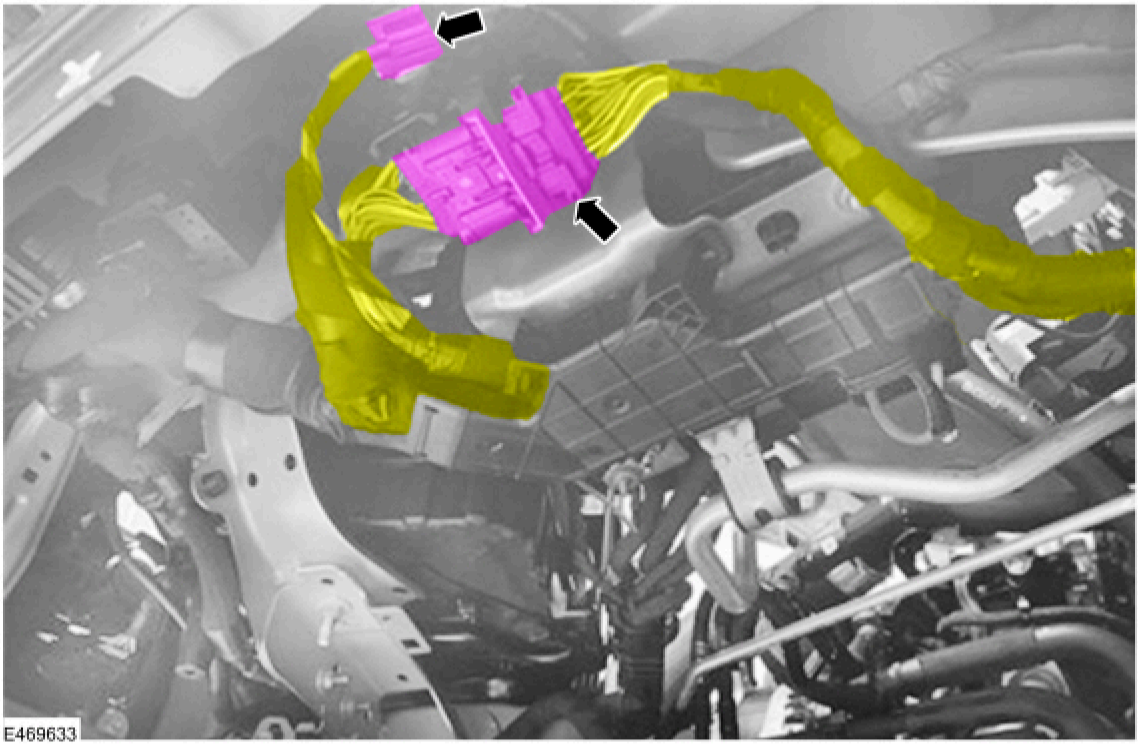
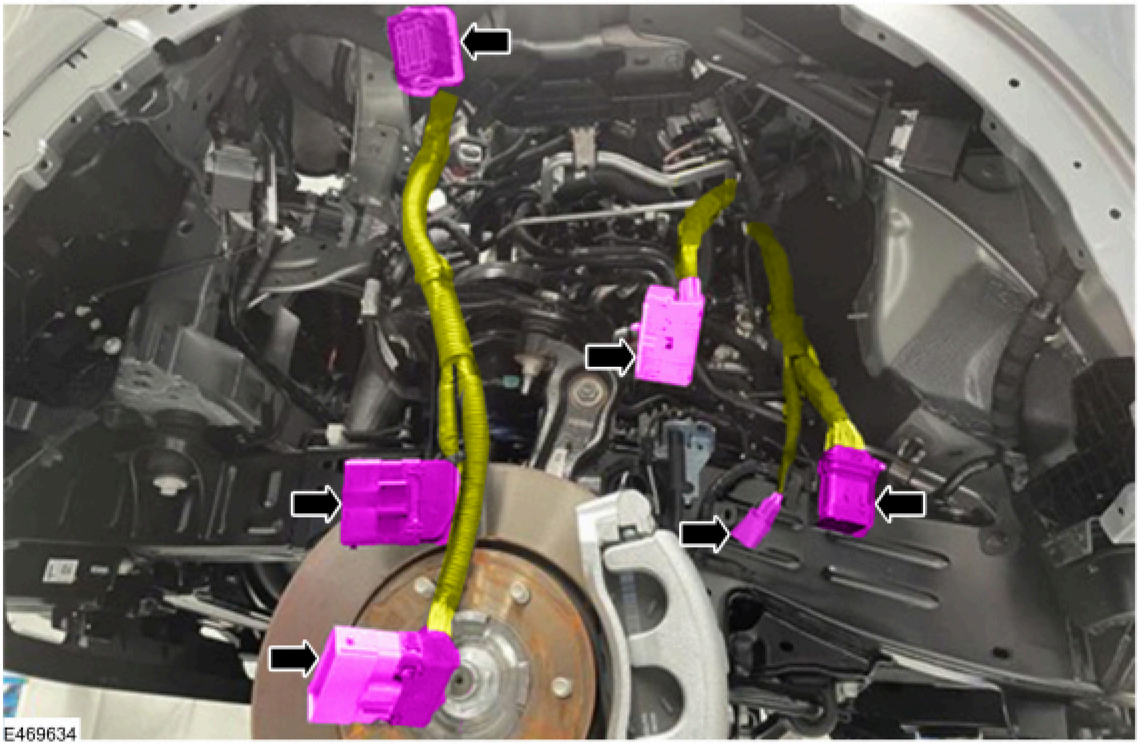


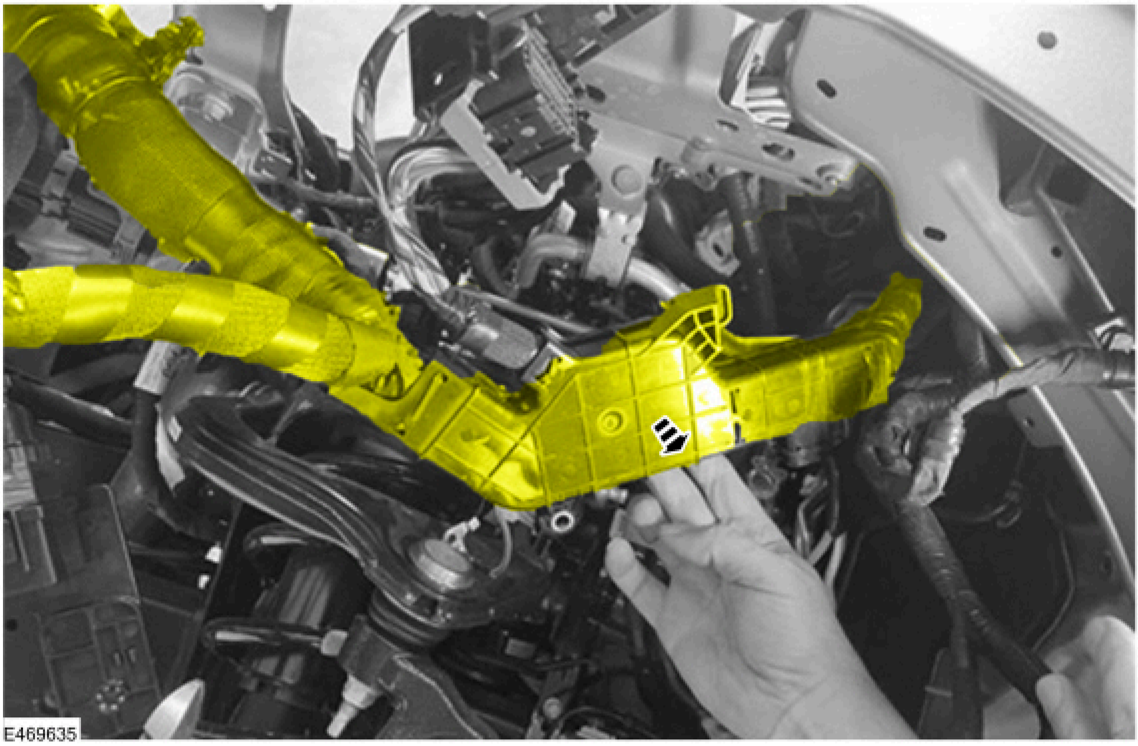
Figure 4



12. Open-up the wiring harness bracket (detach).

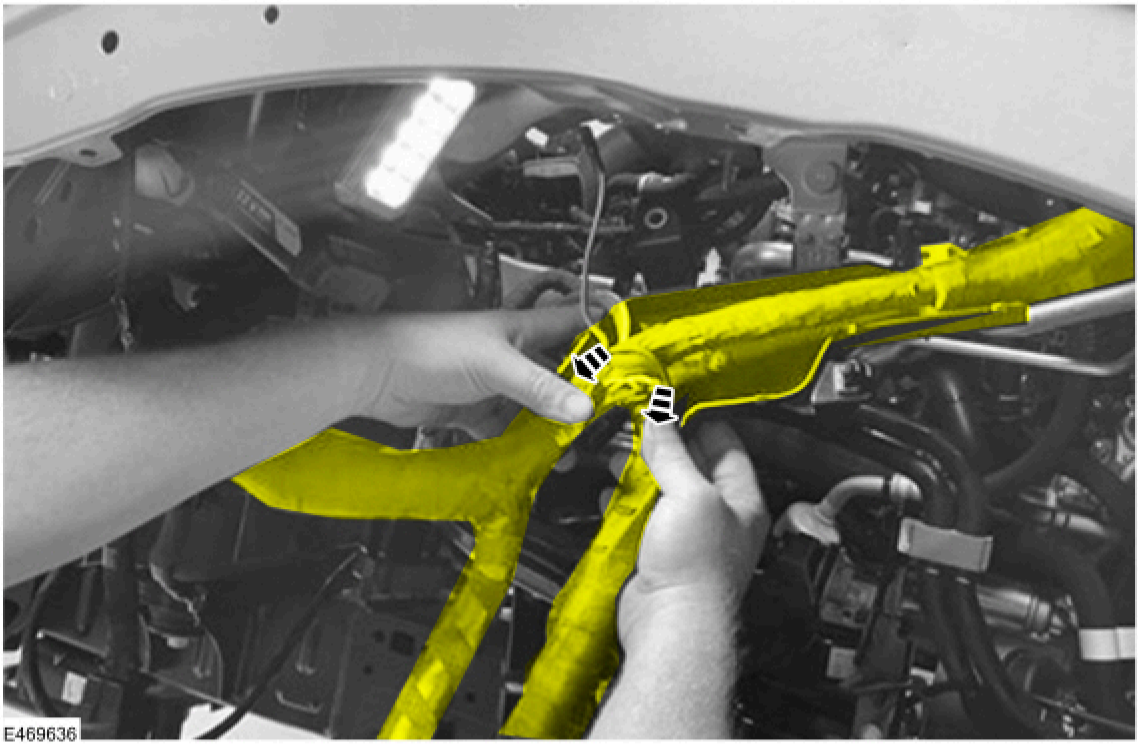
13. Detach the harness above connector C115. (Figure 5)

Figure 5



14. Detach the harness at the 90° bend - take care to release the clips. (Figure 6)

Figure 6

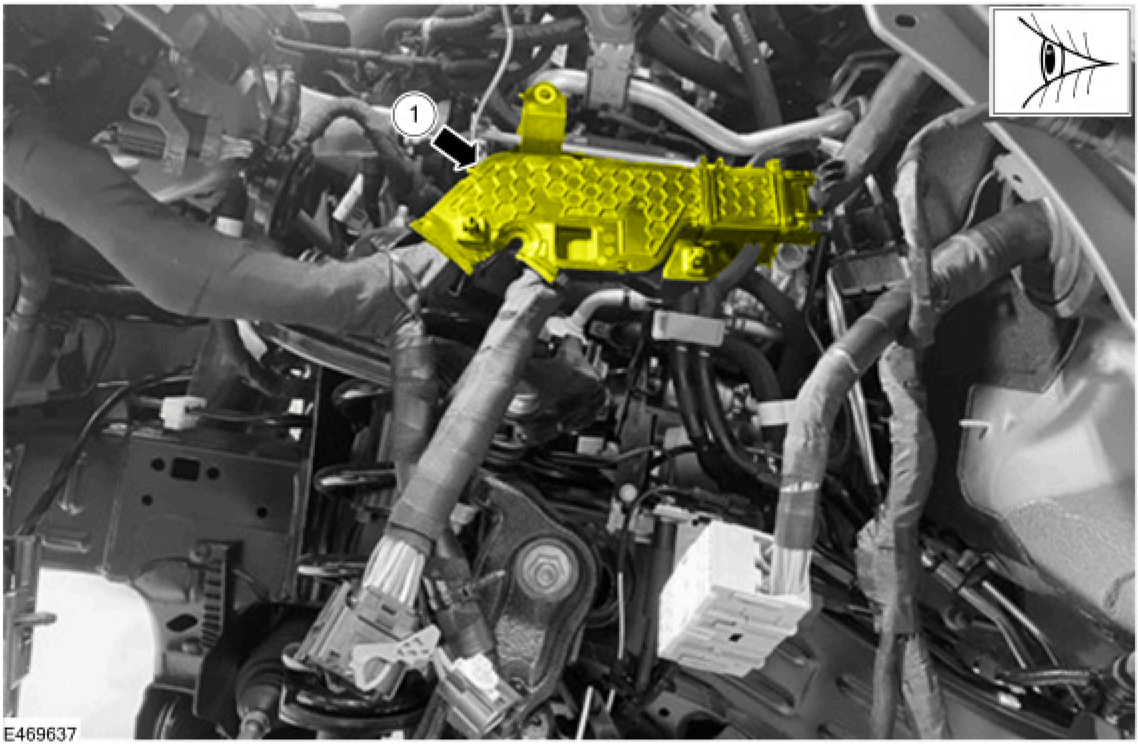


Removal of Existing Harness Protective Coverings in Preparation for the Wiring Pigtail Repair

IMPORTANT: Carefully note the location of all protective coverings, tapes and clips on the wiring harness before proceeding. Taking a few photos of the repair area before proceeding can help to ensure that all protective coverings, tapes and clips are replaced/re-installed into their correct positions upon repair completion. Replacement tapes, protective coverings and cable tie are provided in the TSB pigtail repair kit.

1. Observe and note the location of the black plastic former. Note for reference later that splice joints should be avoided within the zone behind this black plastic former. Splice joints within this zone will make it difficult to refit the black plastic former onto the repaired harness. (Figure 7)

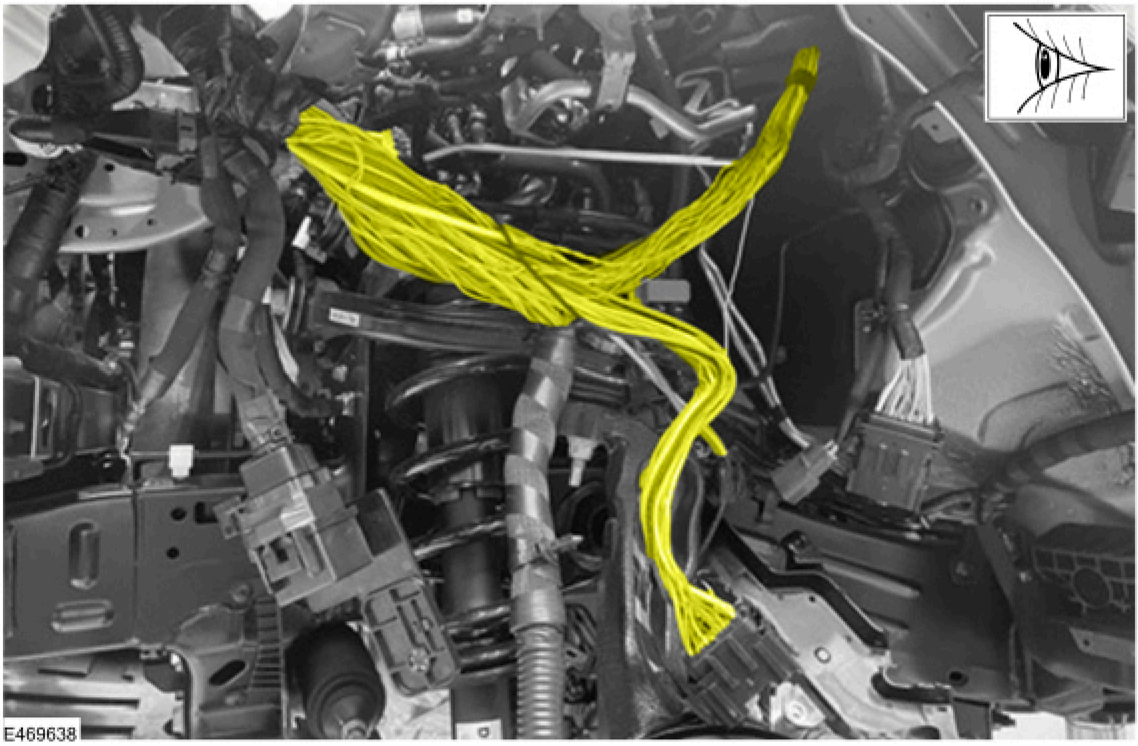
Figure 7



Item	Description
1	Black Plastic Former

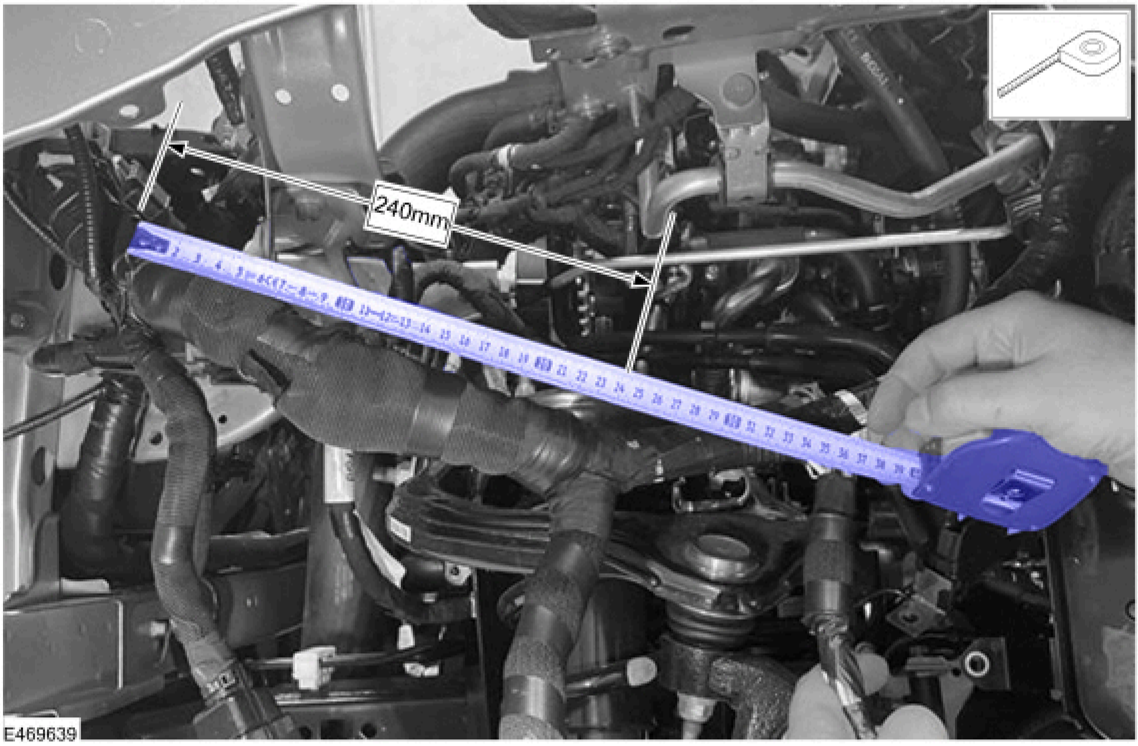
2. Proceed to carefully open-up the wiring harness on the engine wiring harness side (14290) by removing the black plastic former, protective coverings and tape to the extent shown in (Figure 8). Take great care to ensure that individual wires are not damaged when removing the protective coverings.

Figure 8



3. The 240mm approximate dimension shown in (Figure 9) provides a guideline for the extent to which the tape and protective coverings should be removed.

Figure 9



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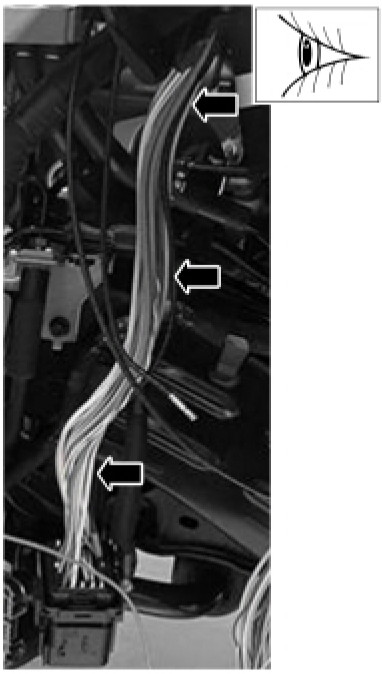
4. Proceed to carefully open-up the wiring harness on the body wiring harness side (14A005) by removing the protective coverings and tape to the extent shown in (Figure 10). Take great care to ensure that individual wires are not damaged when removing the protective coverings.

Figure 10



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5. If the vehicle is fitted with a front camera (feature code J3KAH), remove the front bumper and grille. Refer to WSM Section 501-19.

C115 Pigtail Connector Repair - General Warnings, Cautions and Repair Guidelines

IMPORTANT: The procedure provided below to cut and splice in the C115 connector requires a very careful and methodical approach. Ensure the following instructions are followed during the repair.

- Ensure you to take time to fully read and comprehend each step in the procedure below before starting any cut and splice procedures
- All wiring repairs must follow the instructions in the WSM including PTS, Wiring, Cell 005 together with the instructions provided in this TSB

• Crimp splice joints are used for the wiring repairs covered by this TSB

• The number of circuits running through the C115 connector on any given vehicle varies significantly and is dependent on the vehicle variant. The total number of circuits ranges from approximately 24 to 56 depending on the specific vehicle variant

• If the vehicle is fitted with a front camera (feature code J3KAH), the coax for this camera runs through the C115 connector at pin location 5. On the Body Wiring Harness side (14A005) the coax runs to connector C275 located in the LH front footwell. On the engine bay side (14290) the coax runs to the front camera located in the front grille

• There are a number of cavities in the C115 connector which may have different circuits running through them depending on the specific vehicle variant. The more common variations are shown in Table 1 and 2, however these tables are not exhaustive. The result is that the wire colors and gauges on the service pigtail covered by this TSB may not always match the wire colors and gauges on the vehicle for some cavities. Expect to potentially find 1 or more additional non-matching wires that are not covered in Figure 1 and 2. The pigtail will always have the same or larger gauge wire than the matching circuit on the vehicle

• For the reasons above, it is absolutely critical that only 1 circuit is cut and spliced at a time

• Only use the consumable tapes and coverings supplied as part of this TSB. The consumables provided meet Ford specifications and are critical to the success of the repair

• There may be multiple twisted pairs running through the C115 connector including shielded twisted pairs on some vehicles. Cut and splice of twisted pair circuits must conform to the instructions provided in PTS, Wiring, Wiring and Connector Repair Procedures Cell 005. Crimp joints on twisted pair circuits must should be made within 50mm (2 inches) of the connector. If a repair is required where a twisted pair is present, ensure the twisted pair maintains the same twists per 25mm (inch) after the repair that it had prior to the repair to within 50mm (2 inches) of the connector. Always ensure that the shielding removed or damaged from a twisted pair during the TSB procedure is reinstated to pre-TSB condition using the shielding tape provided in the TSB pigtail kit

• Avoid crimp splices in areas of movement or tight bends

• Stagger the crimp connectors as much as possible. Avoid placing heat shrink crimp connectors next to each other to minimize the increase in wiring harness diameter

• Always leave enough length in the wires to properly insert the stripped ends into the heat shrink crimp connector. Wires that are too short can put excess strain on the wire where it enters the connector, which can cause wiring harness resistance and weak points

• Never attempt to disassemble the C115 connector. Disassembly of the C115 connector can disturb/damage the fragile mat seal and can result in future water ingress and damage to the connector

• Ensure that the radius of any bend in the coaxial cable overlay is NO less than 30mm. A radius of less than 30mm can cause damage to the coaxial cable

Reference Images of C115 Connector Layout (Figure 11 and 12)

Figure 11 – Male connector (body harness side - 14A005)

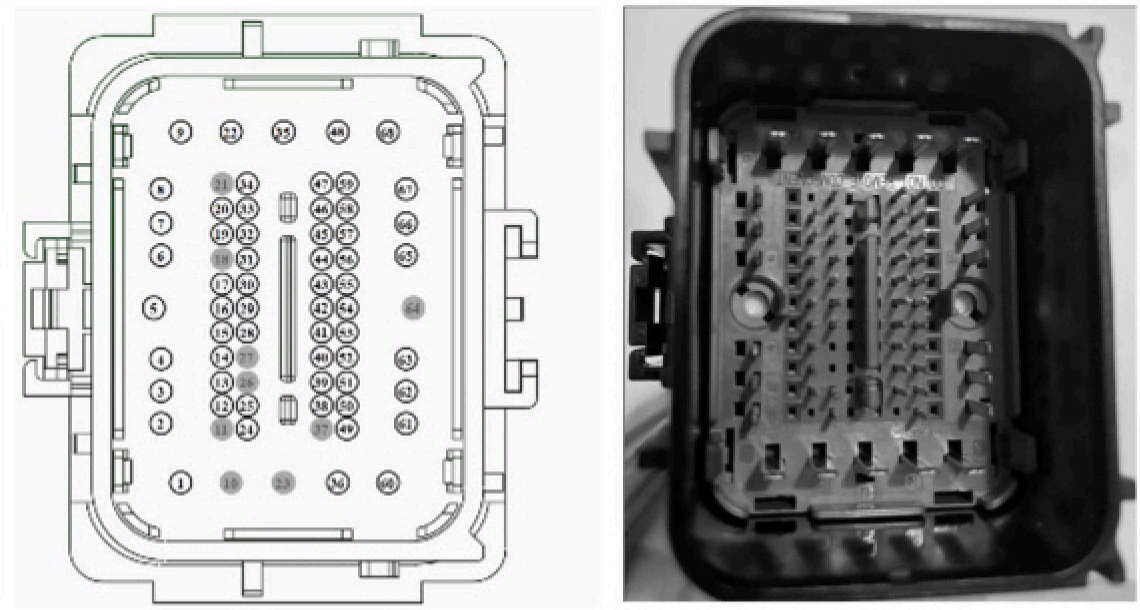

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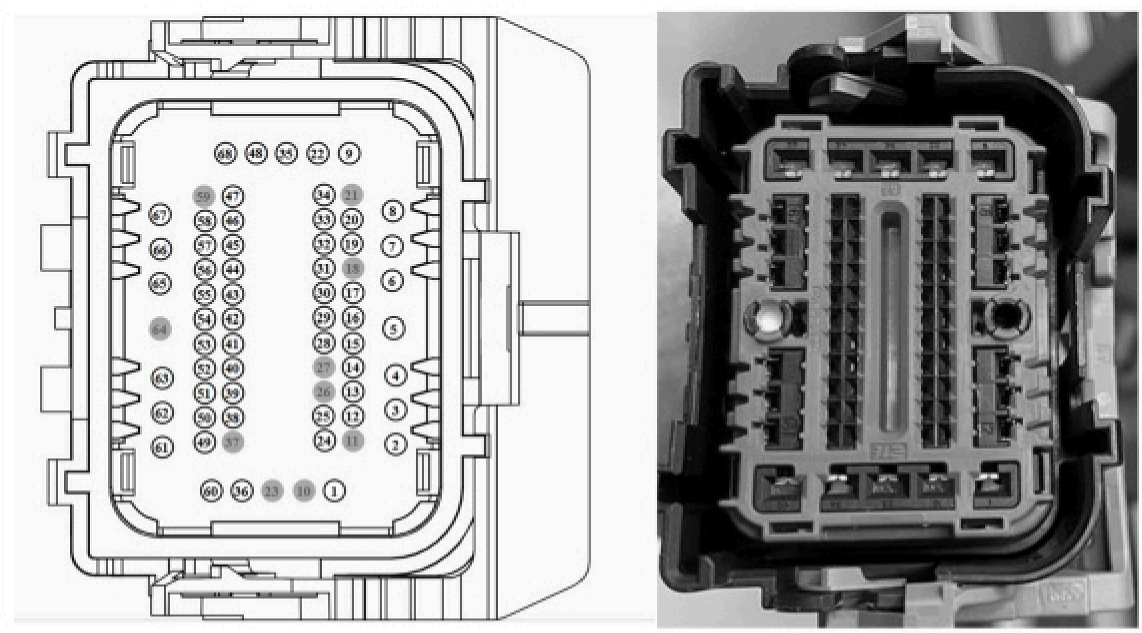
Figure 12 – Female connector (engine bay harness side - 14290)



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Service Pigtail Kit Contents:

RB3Z-14S411-VCA (Table 1)

ITEM	PART NUMBER	PART DESCRIPTION	QTY
1	ESXU5T-14A099-CG03-08	COT TUBE 2950MM (+/-10)	1
2	ESKU5T-1A303-AA(D10N03)	ANTI ABRASION TAPE 25M	1
3	HU5T-14E047-MA	TIE STRAP	22

RB3Z-14S411-HWA (Table 2)

ITEM	PART NUMBER	PART DESCRIPTION	QTY
1	ESXU5T-14A099-CG03-08	COT TUBE 290MM (+/-5)	1
2	ESXU5T-14A099-CG03-08	COT TUBE 830MM (+/-5)	1

ITEM	PART NUMBER	PART DESCRIPTION	QTY
3	ESKU5T-1A303-AA(D10N03)	ANTI ABRASION TAPE 25M	1
4	HU5T-14E047-MA	TIE STRAP	10

RB3Z-14S411-VBB (Table 3)

ITEM	PART NUMBER	PART DESCRIPTION	QTY
1	MU5T-14A464-HD	FEMALE CONNECTOR PIGTAIL	1
2	NU5T-14666-BA	CAVITY PLUG (FOR ADD TO EMPTY CAVITY)	6
3	NU5T-14666-EA	CAVITY PLUG (FOR ADD TO EMPTY CAVITY)	3
4	3U2J-14488-AA	BUTT SPLICE	50
5	3U2J-14488-BA	BUTT SPLICE	9
6	3U2J-14488-CA	BUTT SPLICE	11
7	ESB-M99D56-A2	HEAT SHRINK TUBE SA3-1 (50MM)	55
8	ESB-M99D56-A2	HEAT SHRINK TUBE SA3-2 (50MM)	10
9	ES-KU5T-14A099-BB (DBU028) (80 MM)	CRUSH SHIELD TUBE	1
10	ES-KU5T-1A303-AA(C10Y00)	HIGH TEMP PVC TAPE	1
11	ES-BU5T-1A303-AA (BD)	FOIL TAPE	1
12	ES-KU5T-1A303-AA(D10N03)	HIGH TEMP ANTI ABRASION TAPE	1
13	ID FORD	FORD ID LABEL-1-W	1

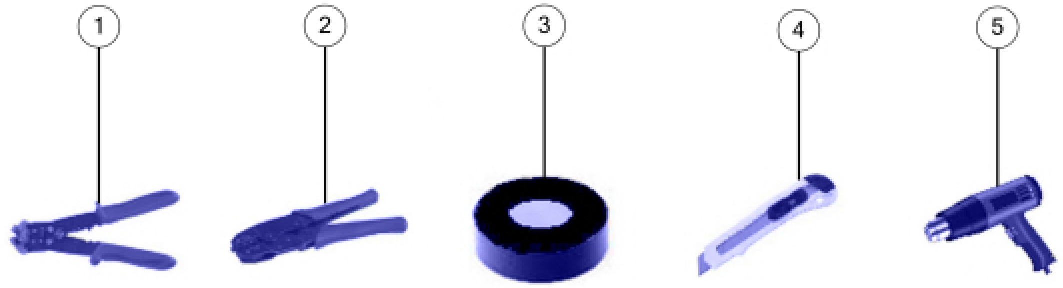
RB3Z-14S411-HVB (Table 4)

ITEM	PART NUMBER	PART DESCRIPTION	QTY
1	MU5T-14A624-CD	MALE CONNECTOR PIGTAIL	1
2	NU5T-14666-CA	CAVITY PLUG (FOR ADD TO EMPTY CAVITY)	6
3	NU5T-14666-FA	CAVITY PLUG (FOR ADD TO EMPTY CAVITY)	3
4	3U2J-14488-AA	BUTT SPLICE	51
5	3U2J-14488-BA	BUTT SPLICE	8
6	3U2J-14488-CA	BUTT SPLICE	11
7	ESB-M99D56-A2	HEAT SHRINK TUBE	65
8	ESB-M99D56-A2	HEAT SHRINK TUBE	7
9	BU5T-14E047-UA	CLIP	1
10	ES-BU5T-1A303-AA (BD)	FOIL TAPE	1
11	ES-KU5T-1A303-AA(C10Y00)	HIGH TEMP PVC TAPE	1
12	ES-KU5T-14A099-BB (DBP028)(180MM)	CRUSH SHIELD TUBE	1
13	F8VB-14A099-KA (80MM)	TWIST TUBE	1
14	ES-KU5T-1A303-AA(D10N03)	HIGH TEMP ANTI ABRASION TAPE	1
15	ID FORD	FORD ID LABEL	1

Recommended tools for the repair:

Suggested Tools for the Repair. (Figure 13)

Figure 13



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Item	Description
1	Wire Stripper
2	Hand Crimp Tool
3	PVC Insulation Tape (tape is provided in the TSB pigtail kit)
4	Cutting Knife
5	Hot Air Gun

Reference Table for Wire Color/sizes on the TSB pigtail Connector vs Potential Wire Color/sizes on the Vehicle:

NOTE: Reference table for wire color/sizes on TSB pigtail connector vs potential wire collar/sizes on the vehicle. (Table 5)

RB3T-14B475-HV* Golden Pigtail (Table 5)

PIN	Pigtail Circuit Size (mm2)	Pigtail circuit color	Possible smallest wire size on vehicle	Possible wire colors on vehicle		
2	0.5	YE/BU	0.5	WH/BN	GN/BU	YE/BU
4	1.5	VT/OG	0.5	GY/BN	BN	VT/OG
6	0.5	BN/VT	0.5	VT/WH	BN/VT	-
18	0.75	GY/OG	0.5	GY/BN	GY/OG	-
30	0.5	VT/WH	0.35	YE	VT/WH	-
32	0.5	YE/VT	0.5	BU/GN	YE/VT	-
33	0.5	GY/BN	0.5	BU/GN	GY/BN	-
38	0.5	VT/OG	0.5	GY/BN	VT/OG	-
42	0.5	VT/GN	0.35	BU	VT/GN	-
43	0.5	GN/OG	0.35	WH	GN/OG	-
56	0.5	YE/GN	0.35	BU	YE/GN	-
58	0.5	YE/VT	0.35	BK/VT	YE/VT	-
59	0.75	GY/BN	0.75	VT/OG	GY/BN	-
60	1.5	BU/GY	1.5	VT/OG	GY/BN	BU/GY
66	1	GY/BN	1	VT/OG	GY/BN	-
68	4	YE/RD	1.5	YE	YE/RD	-

RB3T-14B475-VB* Golden Pigtail (Table 6)

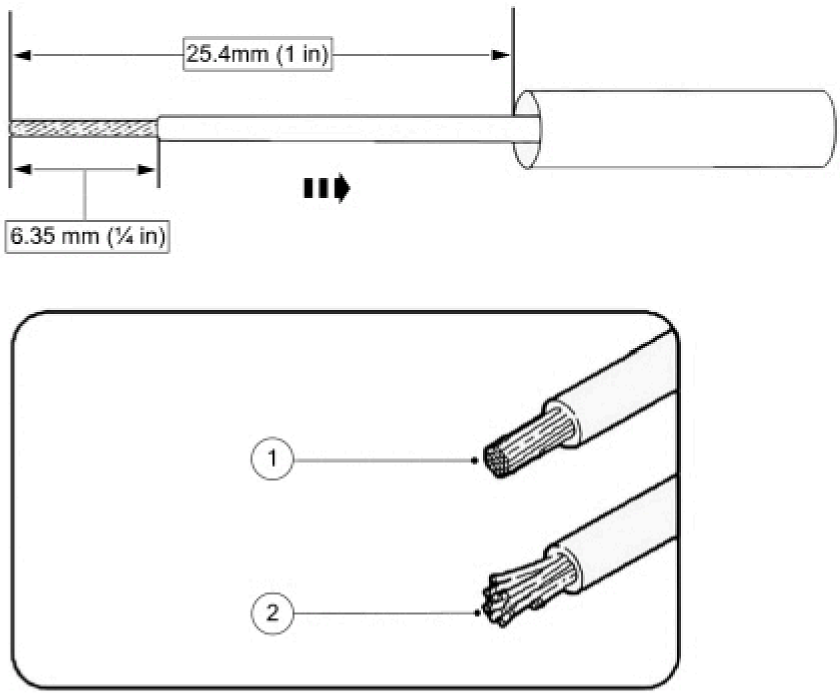
PIN	Pigtail Circuit Size (mm2)	Pigtail circuit color	Possible smallest wire size on vehicle	Possible wire colors on vehicle			
2	0.5	YE/BU	0.25	VT/OG	VT/WH	GN/BU	YE/BU
3	0.5	GN/RD	0.5	VT/OG	GN/BU	GN/RD	-
4	1.5	VT/OG	0.5	BN	VT/OG	-	-
6	0.5	BN/VT	0.5	VT/WH	BN/VT	-	-
18	0.75	GY/OG	0.5	GY/BN	GY/OG	-	-
30	0.5	VT/WH	0.35	YE	VT/WH	-	-
33	0.5	GY/BN	0.5	BU/GN	VT/GN	GY/BN	-
42	0.5	VT/GN	0.35	BU	VT/GN	-	-
43	0.5	GN/OG	0.35	WH	GN/OG	-	-
51	0.5	BU/WH	0.5	GN/WH	BU/WH	-	-
56	0.5	YE/GN	0.35	BU	YE/GN	-	-
58	0.5	YE/VT	0.35	BK/VT	YE/VT	-	-
59	0.75	GY/BN	0.75	VT/OG	GY/BN	-	-
66	1	GY/BN	1	BN/BU	VT/OG	GY/BN	-
68	4	YE/RD	2.5	YE	YE/RD	-	-

NOTE: Tables 5 and 6 are NOT exhaustive lists, refer to TSB section "C115 Pigtail Connector Repair - General Warnings, Cautions and Repair Guidelines" for more information. (Table 6)

Wire Crimping Method

1. Strip 6.35mm (1/4 in) of insulation from each wire end, taking care not to nick or cut wire strands. (Figure 14)

Figure 14



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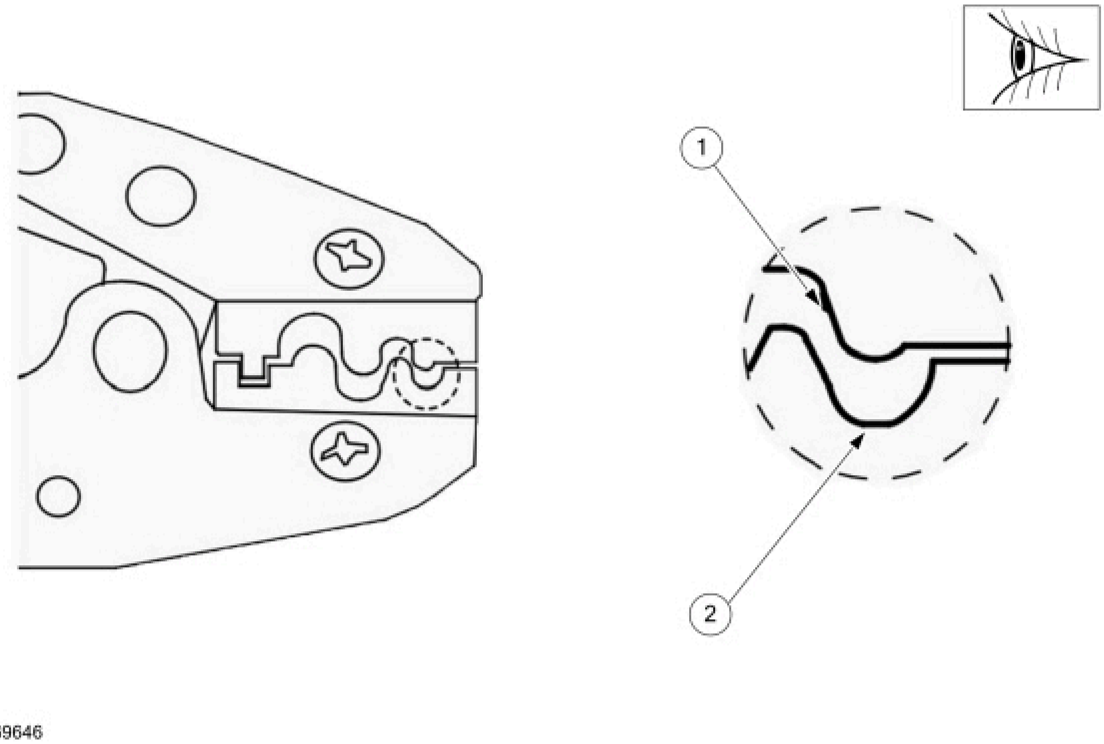
Item	Description
1	Proper Wire Strip
2	Improper Wire Strip

2. Install the heat shrink tubing.

3. Select the appropriate butt splice for the wires to be spliced. (Table 7 below)

4. Identify the appropriate crimping chamber on the hand crimp tool by matching the wire size on the dies with the wire size stamped on the butt splice. (Figure 15)

Figure 15



Item	Description
1	Indenter
2	Cavity

5. Center one end of the wire splice in the appropriate crimping chamber.

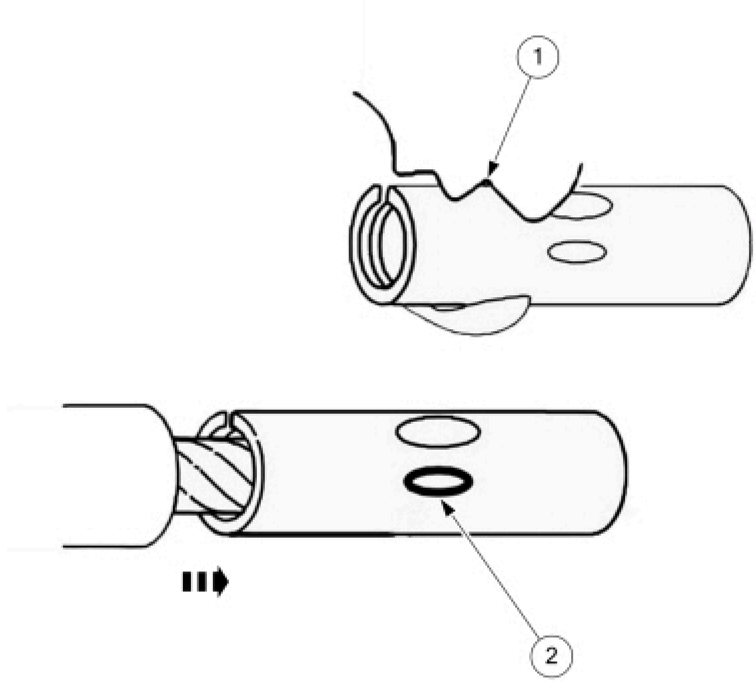
6. If present, be sure to place the brazed seam toward the indenter.

7. Insert the stripped wire into the barrel (butt splice).

8. Holding the wire and barrel in place, squeeze the tool handles until the ratchet releases.

9. Repeating Steps 4-8, crimp the other half of the splice. (Figure 16)

Figure 16



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Item	Description
1	Indenter
2	Wire Stopper

10. Evenly position the supplied heat shrink tubing over the wire repair.

11. Recommendation for 0.35 mm2 wire gauge circuits – Fold the stripped end prior to inserting in a barrel crimp/butt splice to ensure a good quality crimp. (Figure 17)

Figure 17

E469648

12. Perform a gentle pull test after completing each crimp to ensure good integrity of the crimped connection.

13. Use an appropriate butt splice per wire gauge as per the below spec. (Table 7) (Figure 18 to 20)

(Table 7)

Part	Actual Size	Details
330367	22-16 Ga.	Part 330367 is called 18-20 but has been validated per USCAR-21 for wire sizes 0.35mm2 through 1.5mm2 wire (22 through 16).
330368	16-14 Ga.	Part 330368 has been validated per USCAR-21 for wire sizes 1.5mm2 through 3mm2 wire (12 through 16).
330369	12-10 Ga.	Part 330369 can be used for wire sizes 3mm2 through 5mm2 wire (12 through 10).

Figure 18 – Part 330367

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Figure 19 – Part 330368

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Figure 20 – Part 330369

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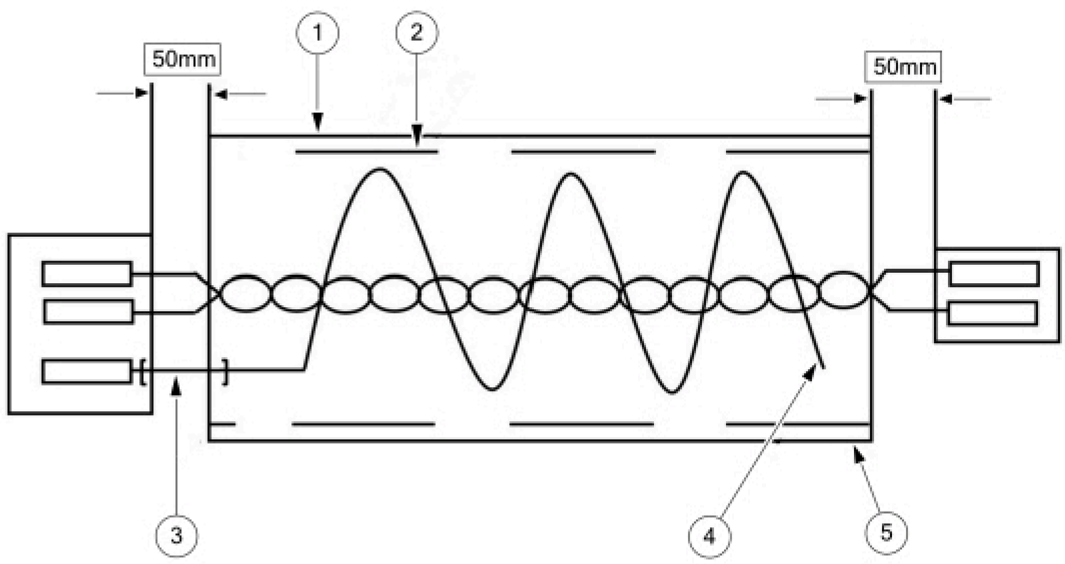
14. Use an appropriate heat shrink per wire size as per the below specification.

(1). SA3-1 – For Circuits with cross section smaller than 4 mm2

(2). SA3-2 – For Circuits with cross section 4 mm2 or bigger

15. Use of shielding for twisted pairs where needed, follow the instructions provided below. (Figure 21)

Figure 21



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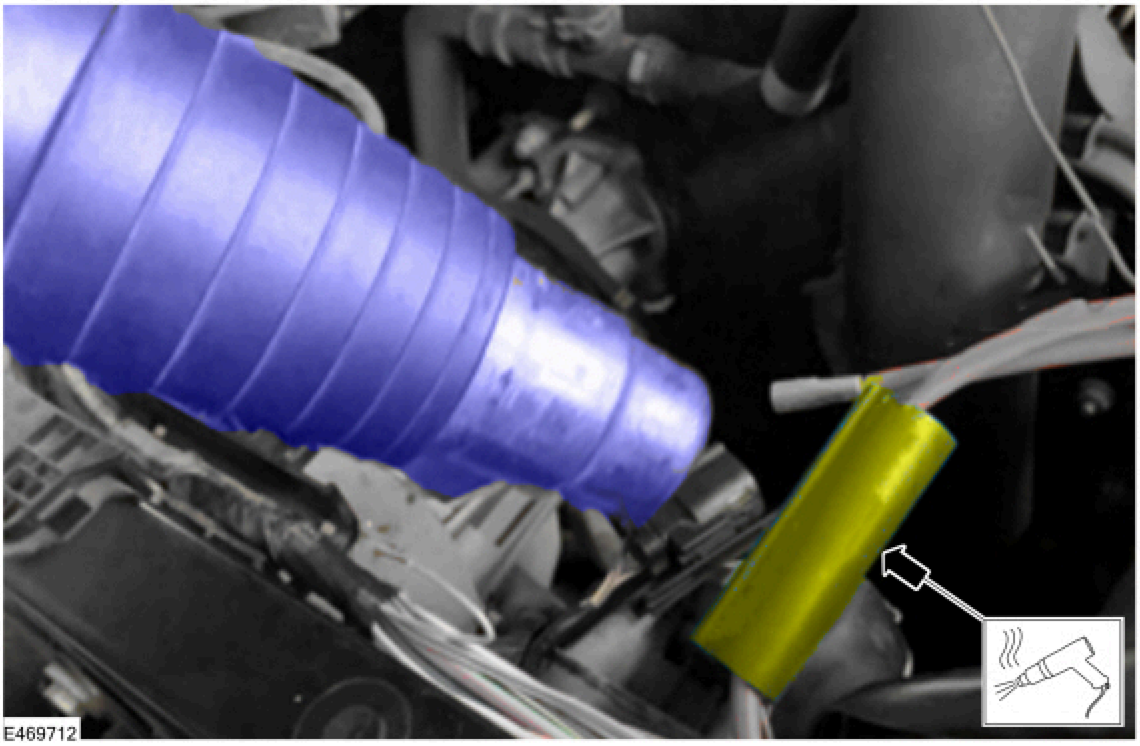
Item	Description
1	B10-00 Or Equivalent
2	ELLIOTT FT1000 Or Equivalent
3	B10-00 Or Equivalent (REF 110mm)

4	Cutting Off
5	Foil Tape Covered With B10-00 Or Equivalent To Be Secured At Both Ends With Sticky Vinyl Tape, The Drain Wires For The Un-therminated Ends, Should End At A Maximum Of 100mm Top Of Terminal And The Foil Covers The Ends Of The Drain Wires

16. Crimp the wiring using the instructions in the Crimping Method.

17. Using a hot air gun with a reflector, heat the shrink sleeve of the crimp connector until the sleeve is tight on the crimp connector and sealant emerges from each end. Continuously move the heat gun to avoid overheating any specific section of the heat shrink tube or wiring. Any discoloration of the wiring insulation indicates overheating. (Figure 22)

Figure 22



C115 Pigtail Connector Installation

1. Read this TSB in full. It is essential that this TSB is read and understood in full before attempting any of the following repair steps.

2. Carefully plan the location of the crimp joint repairs and identify each wire with the approximate location of the crimp joint. Refer back to (Figure 7 above) and ensure that crimp joints are avoided within the zone covered by the black plastic former.

3. Strictly observe and follow all instructions provided in this TSB. It is absolutely critical that only 1 circuit is cut and spliced at a time, refer to the section of this TSB “C115 Pigtail Connector Repair - General Warnings, Cautions and Repair Guidelines” and ensure all instructions are adhered to. Observe all instructions provided in PTS, Wiring, particularly those in Cell 005 - Wiring and Connector Repair Procedures.

4. Install the C115 TSB pigtail connectors RB3T-14B475-HV* and RB3T-14B475-VB* by cutting and splicing only 1 wire at a time.

5. Is the vehicle fitted with a front Advanced Driver Assistance System (ADAS) camera (feature code J3KAH)?

(1). Yes – proceed to Step 6.

(2). No – Checking the DTC memory and erasing of DTCs should be carried out at the end of the repair.

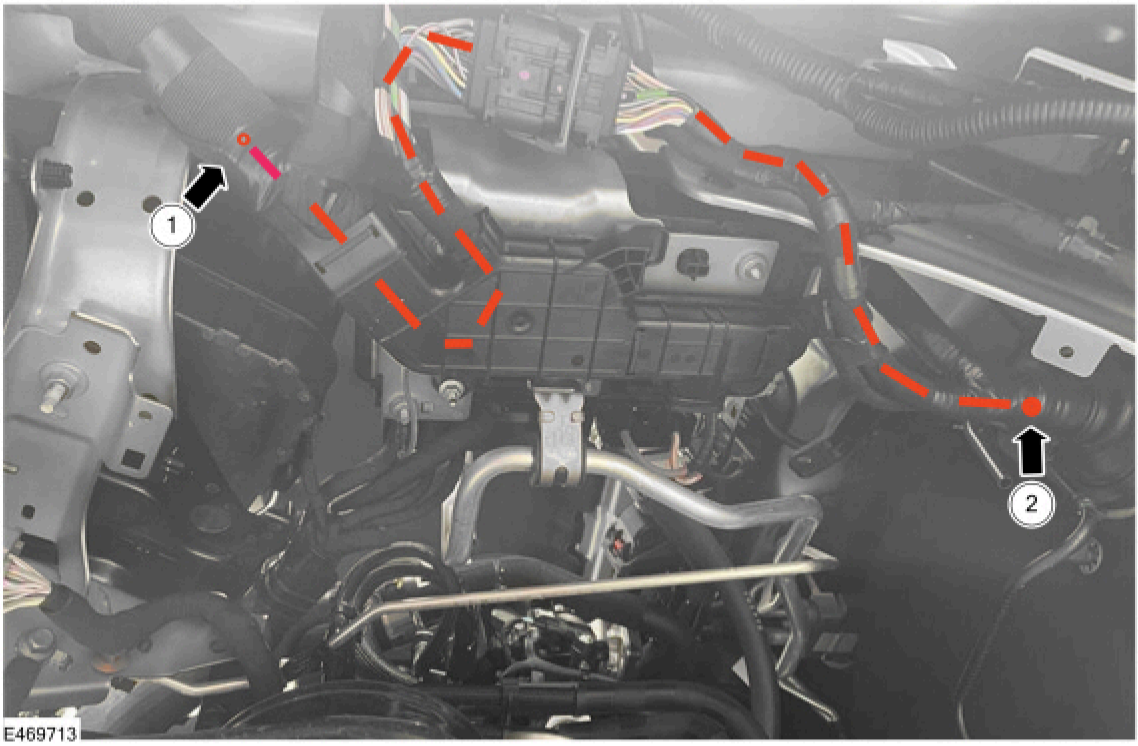
6. Where the vehicle is fitted with a front ADAS camera (feature code J3KAH), then a coaxial overlay will need to be installed. Coaxial overlay part numbers N1WZ-14F662-F and N1WZ-14F662-R are provided with this TSB for this purpose. The coaxial overlay runs through cavity 5 in the C115 connector. The coaxial overlay will be contained within the taped wiring harness bundle for part of it’s length on either side of the C115 connector. Part of the coaxial overlay installation where it is contained within the taped wiring harness will be covered in this section of the TSB and the remainder of the coaxial overlay installation where it is external to the wiring harness bundle will be covered in a later section of this TSB (refer TSB section “Coaxial Overlay Installation”).

7. The section of the coaxial overlay that is contained within the taped wiring harness must be installed at this stage of the TSB together with the installation of the C115 pigtail connectors.

8. Figure 23 below shows the routing of the coaxial overlay either side of the C115 connector. The section highlighted by the orange dotted line is contained within (inside) the taped wiring harness bundle.

- The coaxial overlay breaks out of the taped wiring harness bundle at Callout 1 in Figure 23. This corresponds to Callout 1 in Figure 25 later in this TSB.
- The coaxial overlay breaks out of the taped wiring harness bundle at Callout 2 in Figure 23.

Figure 23



9. After all circuits on the C115 connector are spliced in and the coaxial overlay is installed to the extent shown in preceding Steps 6 to 8 (if fitted) then re-apply all protective coverings and tapes that were removed earlier in this TSB. Refer to the temporary pictures that were taken earlier in this TSB as a reference to ensure that all protective coverings and tapes are reinstalled to their original extent.

Coaxial Overlay Installation

1. Is the vehicle fitted with a front ADAS camera (feature code J3KAH)?

(1). Yes - proceed to Step 2

(2). No – this section of the TSB does not apply. Proceed to TSB Section "Installation of Surrounding Components Post Wiring Pigtail Repair and Post Repair Checks"

2. The coaxial overlay will have been partially installed in the previous section of this TSB, Section "C115 Pigtail Connector Installation". Continue to follow the instructions provided in the following sections to install the coaxial overlay on the Engine bay Side and Body side of the C115 connector.

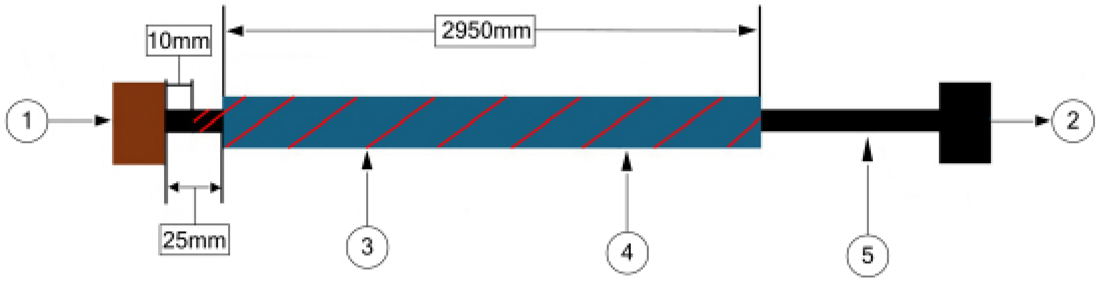
Coaxial Overlay Installation – Engine Bay Side

1. On the engine bay wiring harness (14290) side of the C115 connector, the longer coaxial overlay will be contained within the taped wiring harness bundle until Callout 1 indicated in Figure 23. This will have been partially installed within the taped wiring harness bundle in the previous section of this TSB "C115 Pigtail Connector Installation". The remainder of the coaxial overlay is overlaid externally and is covered in this section of the TSB.

2. On the engine bay wiring harness (14290) side of the C115 connector, the coaxial overlay will be contained within the taped wiring harness bundle until the point (indicated in Figure 23).

3. Figure 24 below shows a schematic layout of the coaxial cable with COT tube and D10-03 tape required for the engine bay (14290) side of the C115 connector from the C115 connector to the front camera.

Figure 24



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Item	Description
1	Camera coaxial male connector inline (Brown)
2	C115 Terminal (coaxial female connector)
3	D10-03 Tape (red cross-hatch)
4	COT Tube
5	Coaxial cable

Layer 1 – COAX (Bottom)

Layer 2 – COT Tube (Middle)

Layer 3 – Overlap D10-03 Tape (Top)

NOTE: COT tube to be installed 25mm away from Camera Connector Inline.

NOTE: D10 overlap tape to cover whole length of Cable + COT except starting 10mm away from Camera Inline Connector and finish at end of COT Tube.

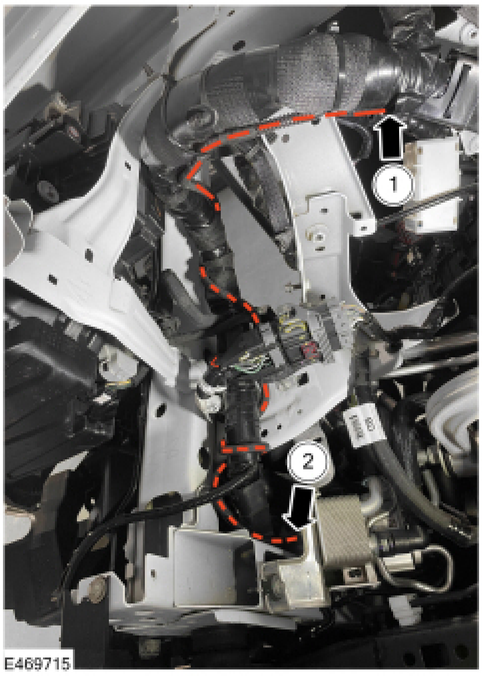
4. Start the overlay coaxial cable overlay repair.

5. Feed the longer coaxial cable into the longer COT tube and apply D10-03 tape according to the instructions and dimensions provided in Figure 24. Ensure that the D10-03 tape overlaps the end of the COT tube near the front camera coaxial connector according to the instructions in Figure 24. Ensure that the D10-03 tape is carefully applied in an overlapping fashion with no gaps for it’s entire length.

6. Working in the LH front wheelhouse, install the coaxial overlay on the engine bay (14290) side following the routing shown in Figure 25. The coaxial cable should be wrapped around the main wiring harness bundle 2-3 times to take up excess length as shown in Figure 25. Secure the coaxial cable to the main wiring bundle using the supplied cable ties. The cable ties must be no further than 150mm apart.

WARNING: Ensure that the radius of any bend in the coaxial cable is NO less than 30mm. A radius of less than 30mm can cause damage to the coaxial cable.

Figure 25



NOTE: Figure 25 Callout 1: The coaxial cable is contained within the main wiring harness bundle starting at Callout 1 in Figure 25 and continuing to the C115 connector. This part of the installation was completed earlier in the TSB procedure.

NOTE: Figure 25 Callout 2: The coaxial cable continues across the front of the vehicle following the main wiring harness bundle in the direction of the arrow at Callout 2 in Figure 25 and continues directly to Callout 2 in Figure 26.

7. Secure the coaxial cable to the main wiring bundle using the supplied cable ties. The cable ties must be no further than 150mm apart.

8. Working on the RH front corner of the vehicle, install the coaxial overlay as shown in Figure 25 following the routing indicated by the dotted line. Secure the coaxial cable to the main wiring bundle using the supplied cable ties. The cable ties must be no further than 150mm apart.

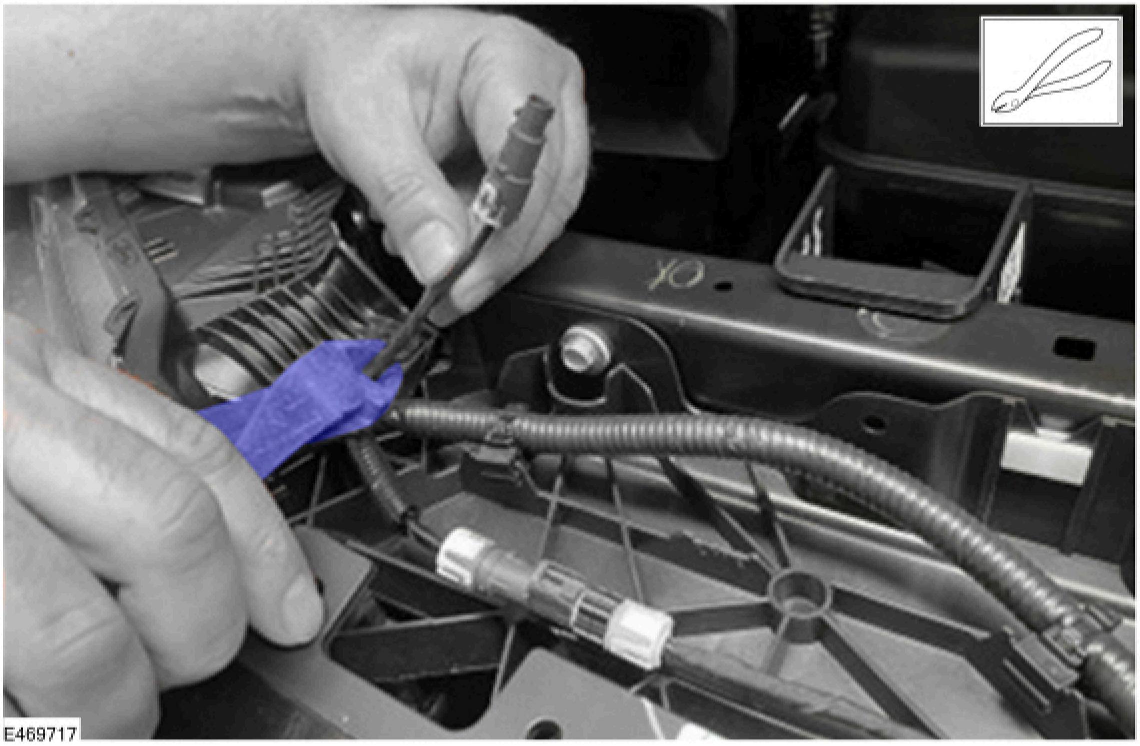
Figure 26

NOTE: Figure 26 Callout 1 - The coaxial overlay continues in the direction of the arrow at Callout 1 in Figure 26 and continues to the front camera.

NOTE: Figure 26 Callout 2 - The coaxial overlay continues across the front of the vehicle following the main wiring harness bundle in the direction of the arrow at Callout 2 in Figure 26 and this continues directly to meet the arrow at Callout 2 in Figure 25.

9. Once the coaxial cable is installed and cable tied in place, the old coaxial cable breakout at the front camera can be cut off as shown in Figure 27. Before cutting off the old coaxial cable ensure the new overlay coaxial cable has the same effective length.

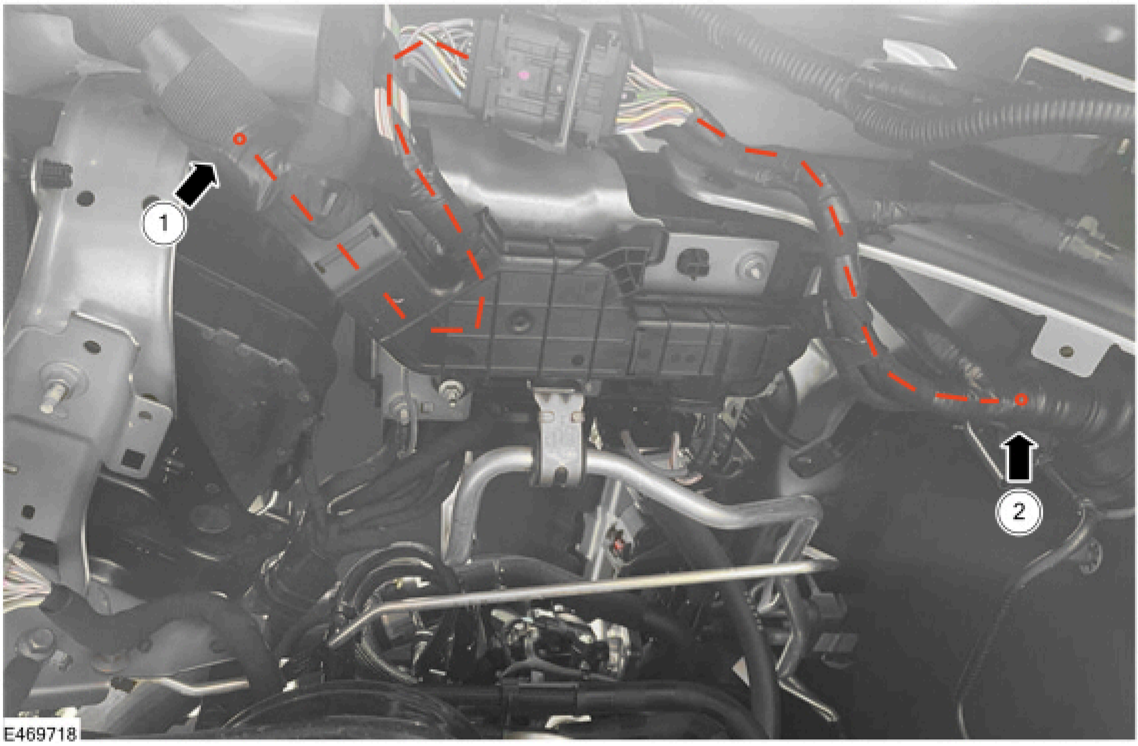
Figure 27



Coaxial Overlay Installation – Body Side

1. On the body wiring (14A005) side of the C115 connector, the shorter coaxial overlay will be contained within the taped wiring harness bundle until Callout 2 in Figure 28. This will have been partially installed within the taped wiring harness bundle in the previous section of this TSB “C115 Pigtail Connector Installation”. The coaxial overlay will break out of the taped harness bundle at Callout 2 in Figure 28 and then will be overlaid externally and pass through the large dash panel grommet. The other end of the coaxial overlay connects into the C275 connector in the LH front footwell area.

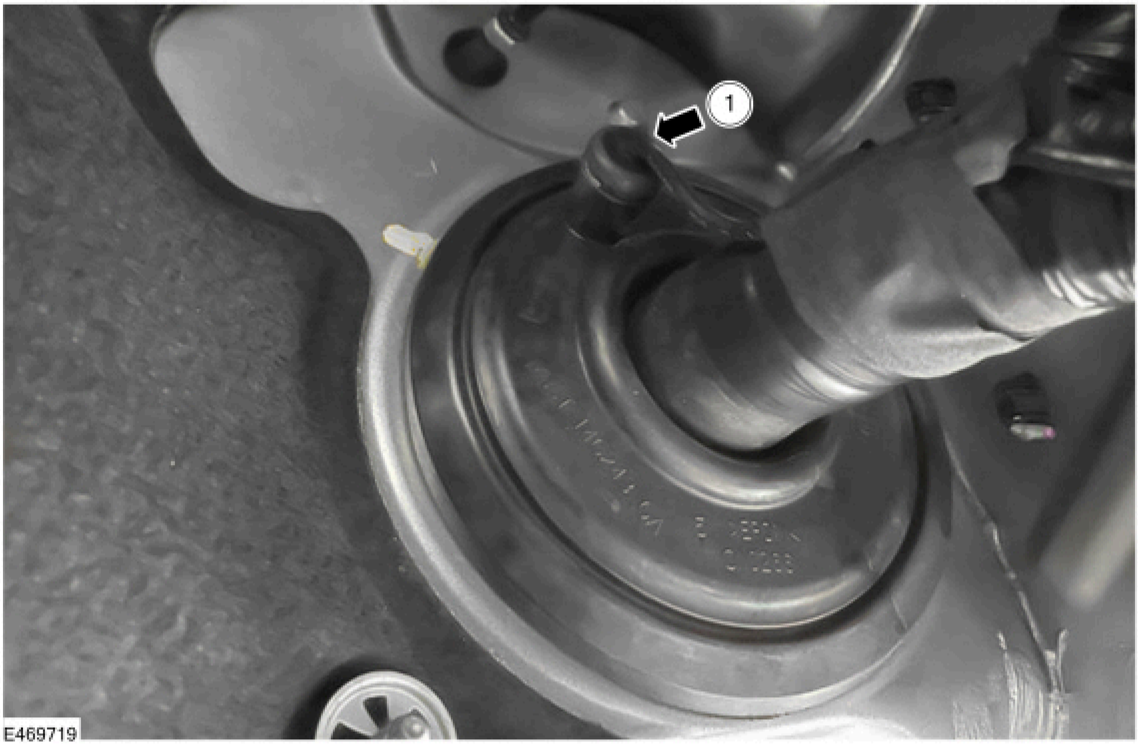
Figure 28



2. The coaxial overlay will break out of the taped wiring harness bundle just prior to the dash panel grommet and will pass through a new hole that will be made in the small service port on this dash panel grommet as shown at

Callout 1 in Figure 29. The short portion of coaxial overlay that runs external to the body wiring harness in the engine bay must be fitted with a section of the plastic COT tube provided to ensure it’s protection

Figure 29



Apply protective covering to the coaxial cable before passing the cable through the grommet on the14A005 side.

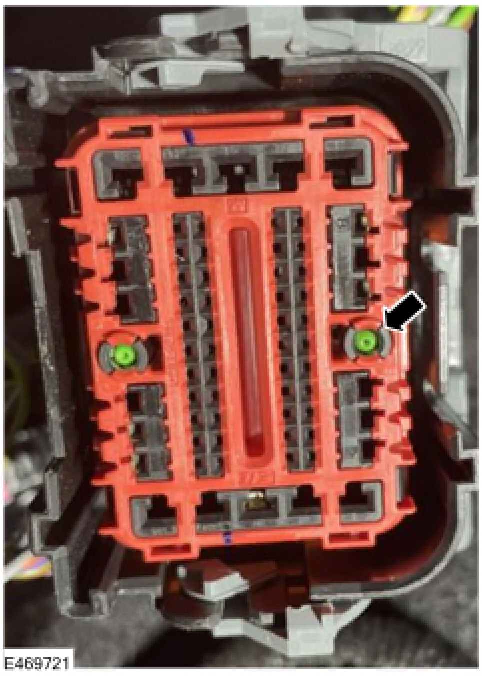
- Create small hole on in the grommet at the dash panel at the service port shown in Callout 1 in Figure 29.
- Apply a short section of COT tube and anti-abrasion tape (D10-03) tape to the exposed portion of the coaxial cable at Callout 1 in Figure 29.
- Feed the coaxial cable through the hole.
- Apply a small amount of silicone sealant on the coaxial cable where its passes through the service port on the dash panel grommet.

3. Remove exiting coaxial cable out from cavity 5 in the C275 connector (circuit VMP52) on the female side only and insert the new coaxial overlay cable into the same cavity (Figure 30 and 31).

4. Secure the coaxial overlay to the existing harness using the supplied cable ties. The cable ties must be no greater than 150mm apart.

Figure 30

Figure 31

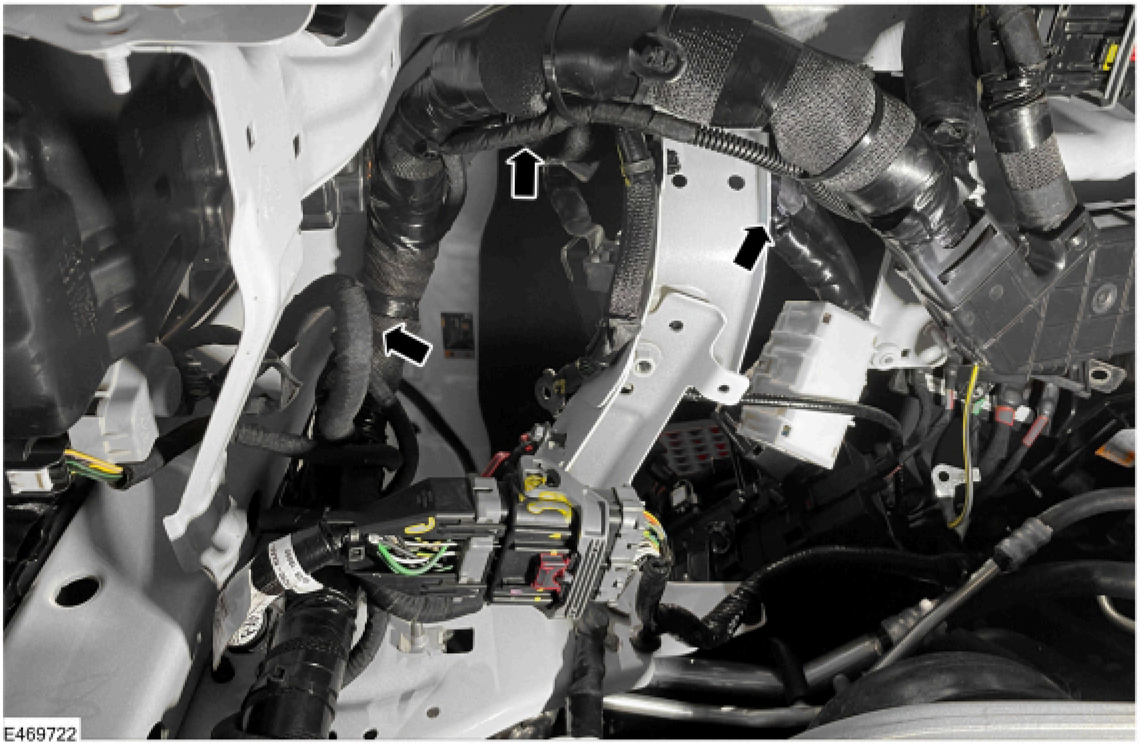


5. Perform a pin-to-pin check for all circuits to establish continuity on all reworked circuits.

6. When the wiring repairs are complete, check the function of the affected electrical components. Always determine what has caused the wiring harness damage (corrosion, sharp body features, faulty electrical components etc.) and make sure the issue has been rectified

7. Insulate the repair area using fabric tape and refit any conduit/trucking. The repaired wiring harness must always be routed and mounted exactly as it was before the repair. (Figure 32)

Figure 32



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Installation of Surrounding Components Post Wiring Pigtail Repair and Post Repair Checks

1. Install all surrounding components removed during the “C115 Connector Inspection” and “Removal of Surrounding Components in Preparation for the Wiring Pigtail Repair” Sections above. Reverse the removal procedures.

2. Connect the Ford Diagnosis and Repair System (FDRS) and check for DTCs. Check to confirm that no new DTC’s are present compared to those that were noted at the very start of this TSB. If any new DTC’s are present then thoroughly inspect the integrity of all wiring repairs to ensure that these are not causing the new DTC.

3. Evaluate the vehicle to ensure the symptoms observed that are covered by the symptom and DTC list of this TSB have now been resolved. Confirm that the vehicle symptoms and DTC’s recorded at the very start of this TSB have now been resolved.

4. Have the applicable vehicle symptoms and DTCs now been resolved?

(1). Yes – TSB complete.

(2). No – Refer to the WSM for further diagnosis/repair.
